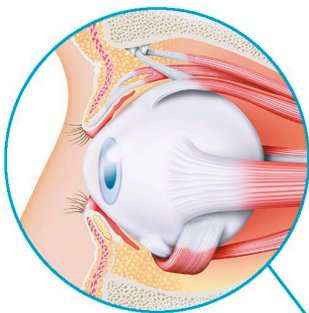


## THE RETICULAR FORMATION

The reticular formation is a structure containing various clusters of neurons (nuclei) together with a series of long, slim nerve tracts that are located in much of the length of the brainstem (see p.93). Its fibres extend to the cerebellum behind, the diencephalon above, and the spinal cord below. The reticular formation comprises several distinct neural systems, each with its own neurotransmitter (the chemical that

passes on nerve signals at the tiny junctions, or synapses, between neurons). One of the reticular formation's many functions is to operate an arousal system, known as the reticular activating system (RAS), that keeps the brain awake and alert. The reticular formation also includes the cardiorespiratory and respiratory centres that control heart rate and breathing, and other essential centres.



### VISUAL IMPULSES

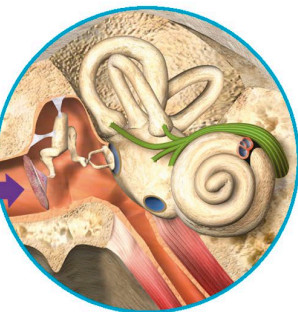
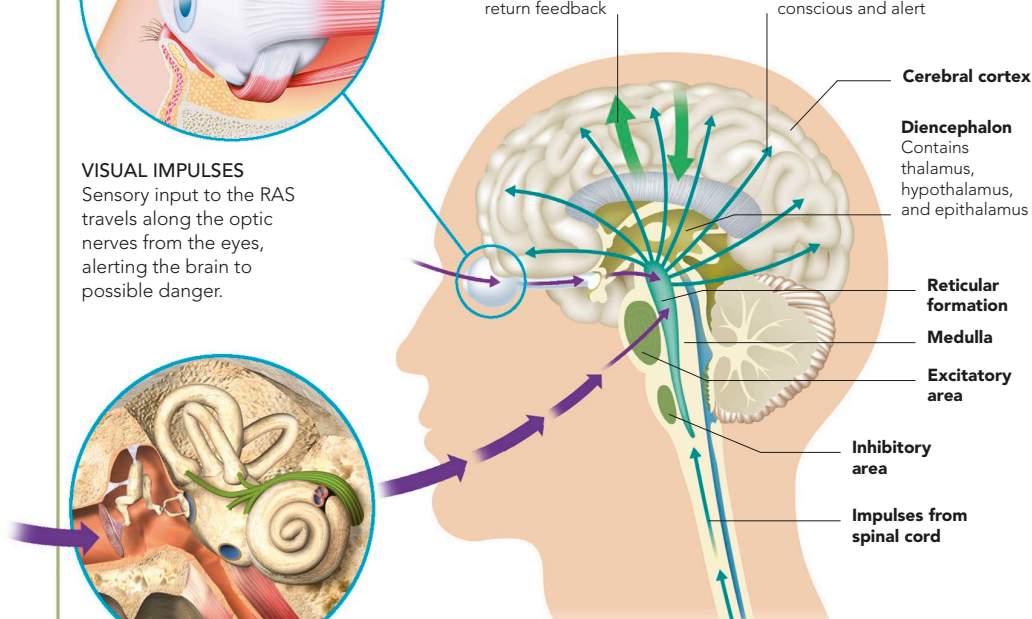
Sensory input to the RAS travels along the optic nerves from the eyes, alerting the brain to possible danger.

### Radiating signals

The RAS sends activating signals up through the midbrain to regions of the cerebral cortex, while other nerve fibres return feedback

### Activating signals

Nerve signals arrive at the cerebral cortex to maintain the state of wakeful readiness so that the mind remains conscious and alert



### AUDITORY IMPULSES

The RAS filters out insignificant sensory information, such as background noise, and reacts if there is a change in input.

### THE RETICULAR ACTIVATING SYSTEM

Long, slender pathways of nerve fibres in the reticular formation, which is located within the brainstem, detect incoming sensory information from many sources. They send activating signals to the higher centres of the brain.